AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter
Monitoring & Health Effects

Issue 2026-11-10 | Entry window 10 November 2026 (full day) | 26 records screened in scope

26 RECORDS IN SCOPE (66 EXCLUDED)

8 OF 10 SUBTOPIC BANDS POPULATED

9 EFFECT ESTIMATES WITH A CI

3 TIER-A RECORDS

4 RECORDS CARRIED ON METADATA ALONE

Signal of the day

In 55,223 hemodialysis patients across California, Oregon and Washington, a smoke plume over the clinic raised same-day mortality (RR 1.19, 1.11–1.27), but plume plus an extreme heat event gave RR 1.92 (1.71–2.17) against 1.02 (0.95–1.10) for plume alone; per 10 μg/m^3 of wildfire PM$_{2.5}$ the mortality RR was 1.03 (1.02–1.06).

University of Maryland, *Environmental Epidemiology*; Fresenius Kidney Care records 2010–2018, time-stratified case-crossover, conditional quasi-Poisson, exposures assigned to the dialysis facility. **Where it is weak:** exposure is clinic-sited, not residential, and wildfire PM$_{2.5}$ is a modelled split of total PM$_{2.5}$; the plume-alone null with a large joint effect means the headline plume signal is mostly heat-coincident, and the abstract gives no wildfire-PM$_{2.5}$ $\times$ heat interaction on the concentration scale. **Why it matters here:** dialysis patients are at the same address three times a week. A low-cost monitor on each clinic is an exact exposure point for one of the most smoke-vulnerable populations, and it would replace the modelled split with a measured one.

What changed today

- **Nitrate leads a component mixture.** Pearl River Cohort, 182,009 participants: each quartile of a six-component PM$_{2.5}$ mixture index raised respiratory mortality (HR 1.29, 1.23–1.34), nitrate weight 0.48, then organic matter and chloride. Single-component HRs of 1.42–2.75 come without increments in the abstract. The same day, a Peking group's global HONO reconstruction names HONO change as a major driver of particulate nitrate trends in Beijing and Delhi.
- **An AI air-quality model beats CAMS at stations but goes negative.** Independent evaluation of the released Aurora 0.4° checkpoint, 96 initialisations in 2024: MSE skill over persistence 38.1–42.6 % for PM$_{2.5}$ and 49.1–52.3 % for PM$_{10}$, station MAE below CAMS in all six region–pollutant pairs, yet 6.27 % of PM$_{2.5}$ and 7.84 % of PM$_{10}$ grid values negative at +12 h and no forecast distribution. Background stations only (preprint).
- **Two cohort nulls and one unscaled positive.** Brigham MS cohort (3,040, mean PM$_{2.5}$ 5.8 μg/m^3): PIRA HR 1.01 (0.96–1.06), EDSS $\geq$ 6 HR 1.03 (0.94–1.13). UK Biobank substance use disorder: PM$_{2.5}$ HR 1.111 (1.081–1.142), PM$_{10}$ 1.070 (1.054–1.086), increment not stated in either abstract, so neither can be pooled.
- **Sensing side is all modelling.** Four sensing records today — Aurora evaluation, size-resolved UFP forecasting in Shenzhen (best NRMSE 60.2 % at 10 nm vs 13.4–14.8 % at 19–100 nm), desert AOD forecasting, and a denoising neural-process interpolator that cut RMSE up to 30 % on PM$_{2.5}$ from real-world air-quality sensors (preprint, network unnamed). No co-location or calibration record.
- **Exposure basis (x1).** One record rests on a low-cost sensor network (the neural-process interpolation benchmark, sensing side) and none on a personal monitor; two rest on regulatory monitors (urolithiasis case-crossover, German ED protocol). On the health side, of 11 records with a recoverable basis 36 % are measured (mostly laboratory exposures) and 64 % modelled or satellite-fused; *every one of today's nine effect estimates rests on modelled exposure* (wildfire split, component surfaces, LUR, satellite zip-code means).

Window continuity

The 9 November issue closed with `last_entry_date` at 9 November. This issue was built on the **12 November 23:38 UTC (17:38 CST) run**. Before 22:00 CST the newest buildable day is yesterday, so this run builds **10 and 11 November as two issues**; **10 November is covered in full**. Tuesday: no rollup is owed; **W46 (8–14 November) falls on Saturday 14 November**. A cross-day pass over the 10–11 November candidates assigned each record to its earliest day and removed 12 records already seen or rejected. No BACKFILL records today. OpenAlex, alphaXiv, bioRxiv/medRxiv and Wiley Scholar Gateway connectors were not available on this session; the OpenAlex harvester leg returned HTTP 429.

Corpus analytics

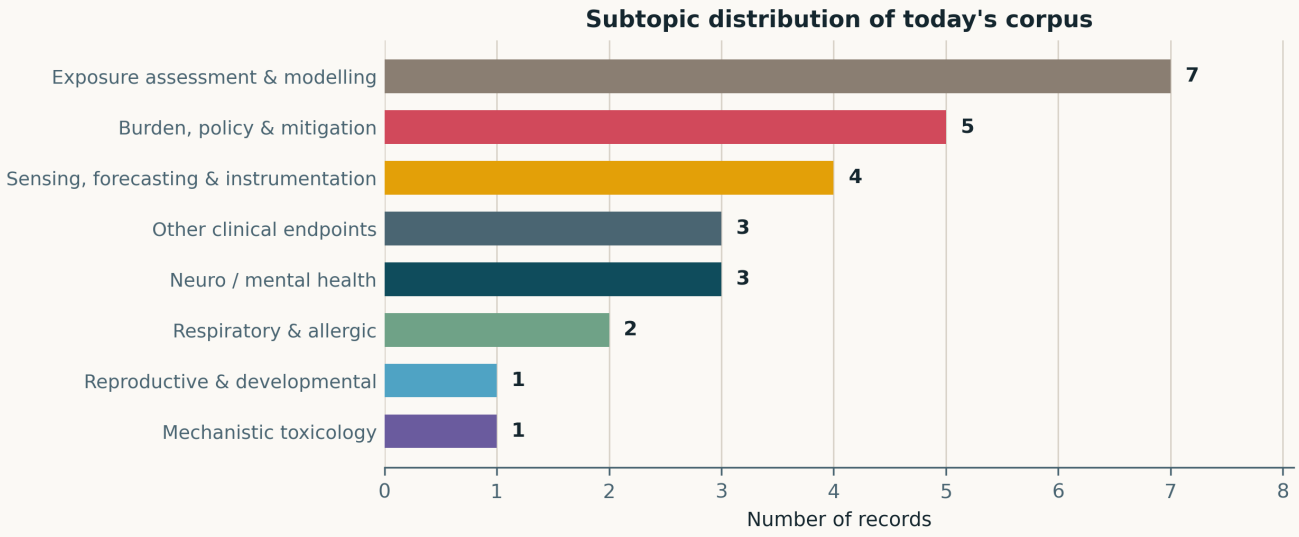


Fig. Subtopic assignment of the 26 records across eight of ten bands: exposure assessment 7, burden 5, sensing 4, other clinical 3, neuro 3, respiratory 2, and one each for reproductive and mechanistic. Cardiovascular and occupational are empty today.

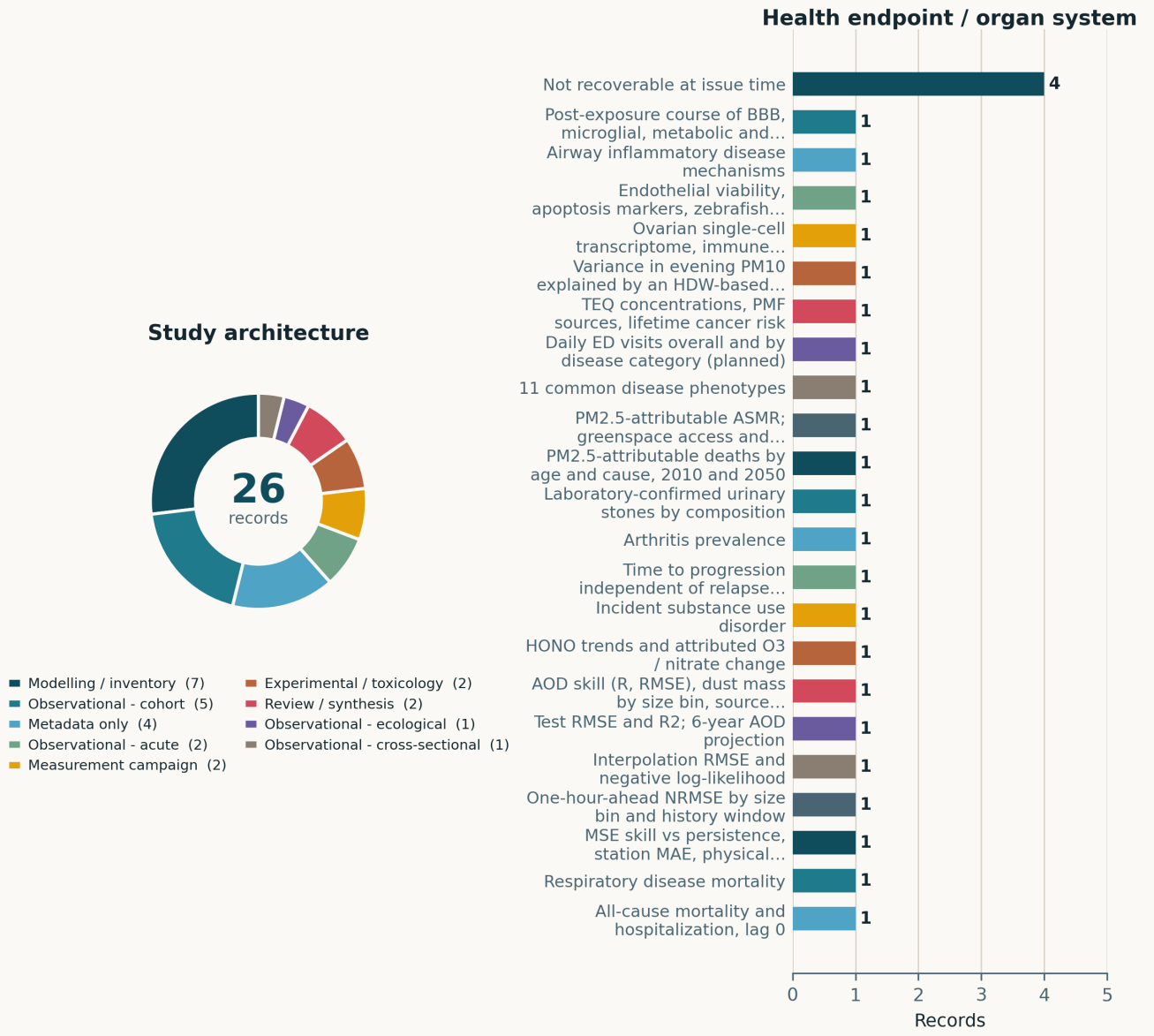


Fig. Left — architecture: modelling 7 (all four sensing records among them), cohort 5, metadata-only 4, and two each for acute designs, measurement campaigns, experiments and reviews, plus one ecological and one cross-sectional record. Right — every recoverable endpoint is a singleton; the bar of 4 is the metadata-only records, and eight endpoints are measurement or forecast metrics rather than clinical outcomes.

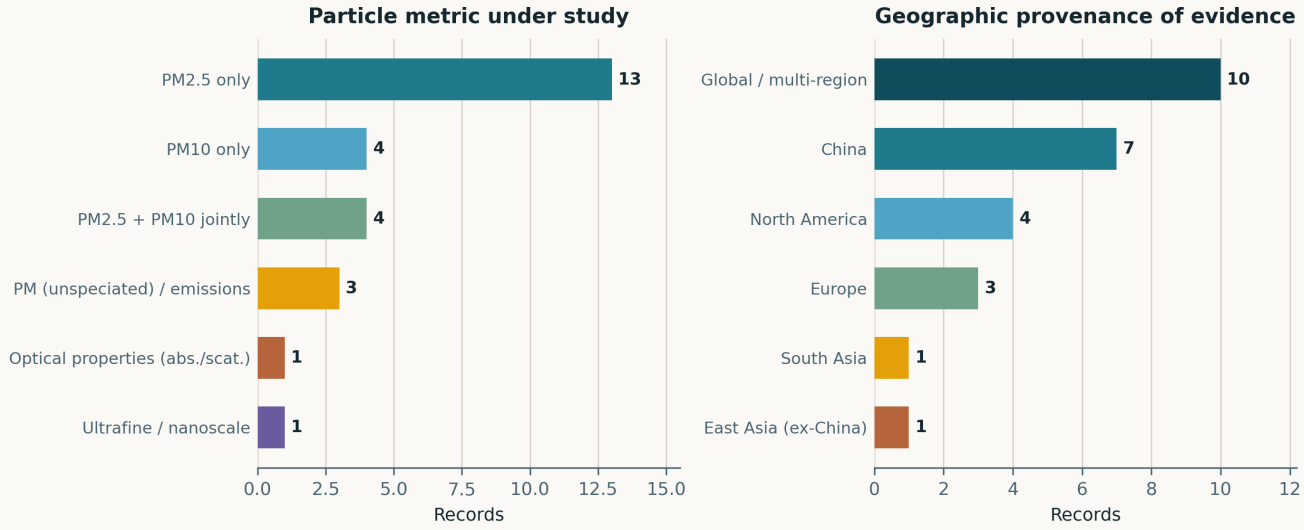


Fig. Left — particle metric: $PM_{2.5}$ alone 13, PM_{10} alone 4, $PM_{2.5}$ and PM_{10} jointly 4, unspeciated PM or emissions 3, and one each for ultrafine number and optical (AOD). Right — geography by region: global or multi-region 10 (including the two multi-country Asian records and two with no stated location), China 7, North America 4, Europe 3 (Italy, Germany, UK), and one each for East Asia ex-China (South Korea) and South Asia (India).

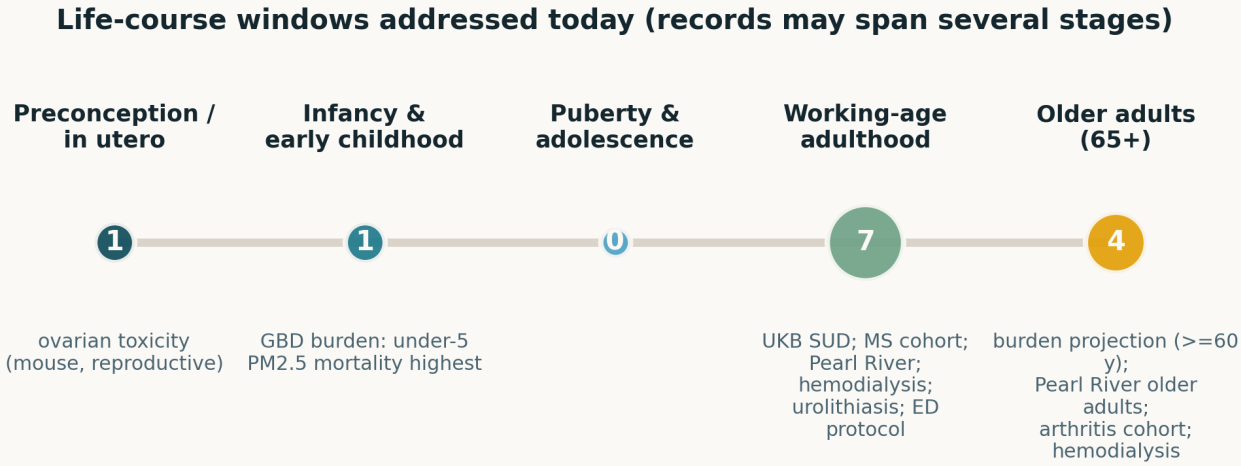


Fig. Life-course windows. Preconception is the mouse ovarian study; early childhood the GBD analysis, where $PM_{2.5}$-attributable mortality is highest under age 5. Working age carries seven records (UK Biobank, MS cohort, Pearl River, hemodialysis, urolithiasis, arthritis cohort, ED protocol); older adults four (2050 burden projection, Pearl River, arthritis cohort, hemodialysis). No adolescent record.

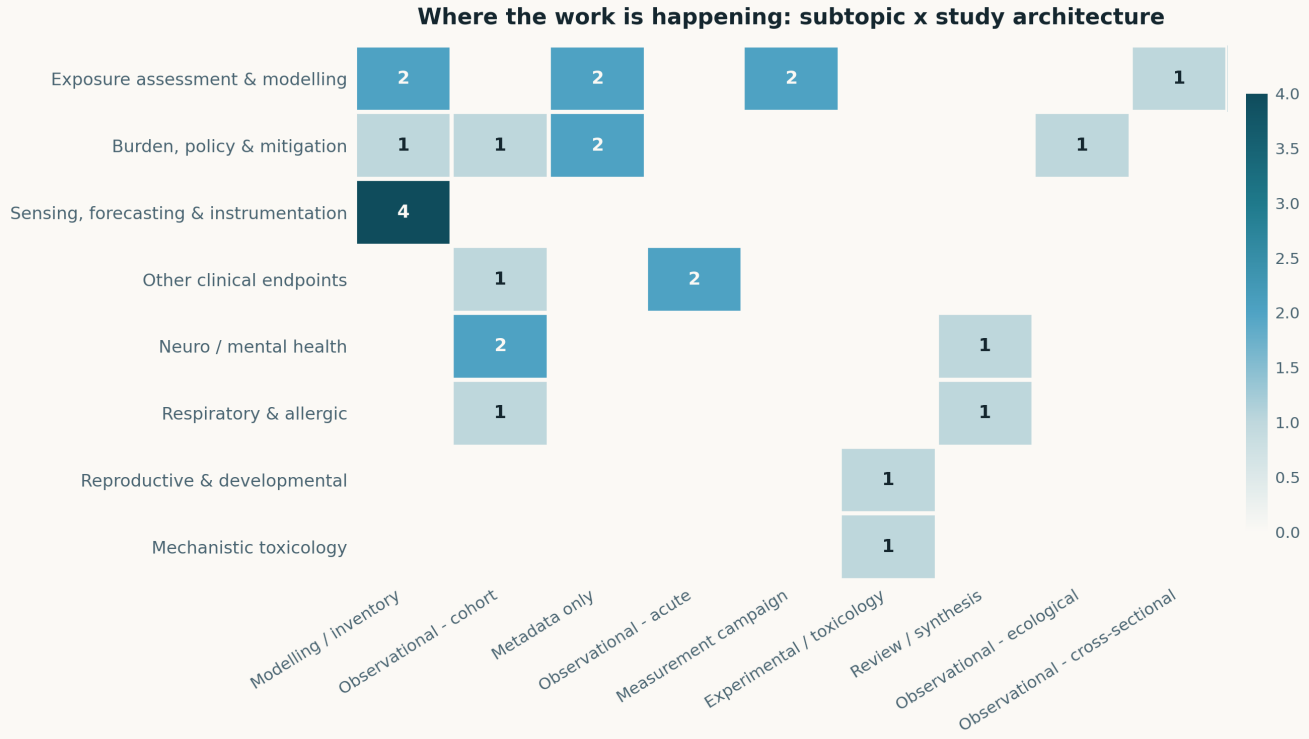


Fig. Subtopic × study architecture. Sensing is entirely modelling (4); exposure assessment splits across modelling (2), metadata-only (2), measurement campaigns (2) and the GIS exposome study (1); burden mixes modelling, the ED protocol (cohort), two metadata-only records and the GBD greenspace analysis. Both acute designs are other-clinical (hemodialysis, urolithiasis).

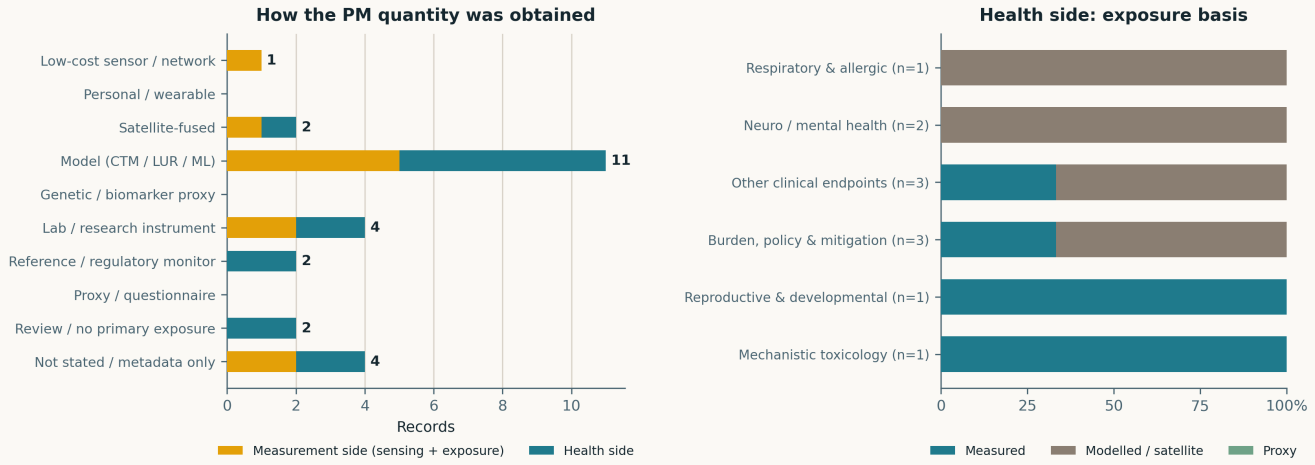


Fig. How each record obtained its PM quantity, classified from the exposure field. 1 of 26 records rest on a low-cost sensor or sensor network; 2 on reference or regulatory monitors. Health side, where an exposure basis is recoverable (n=11): measured 36 %, modelled or satellite-fused 64 %, proxy 0 %. Right panel sorted by modelled share: the subtopics at the bottom are where a measured low-cost network exposure would add most.

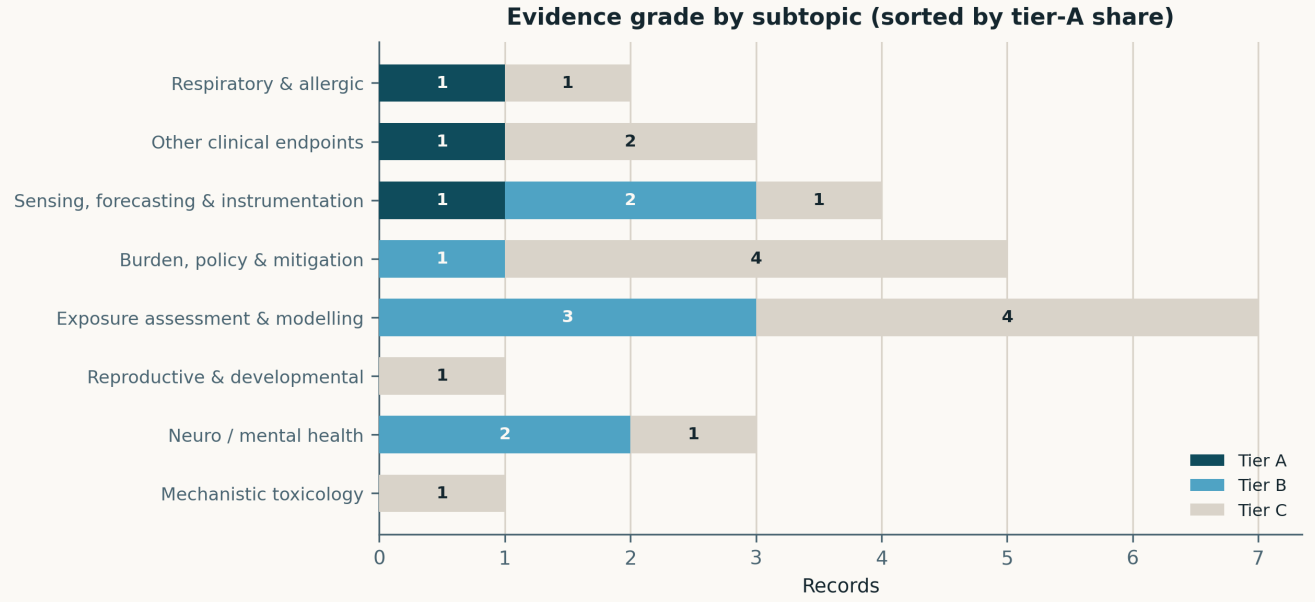


Fig. Tier mix per subtopic. 3 of 26 records are tier A (12 %). Highest tier-A share: *Respiratory & allergic* (1 of 2).

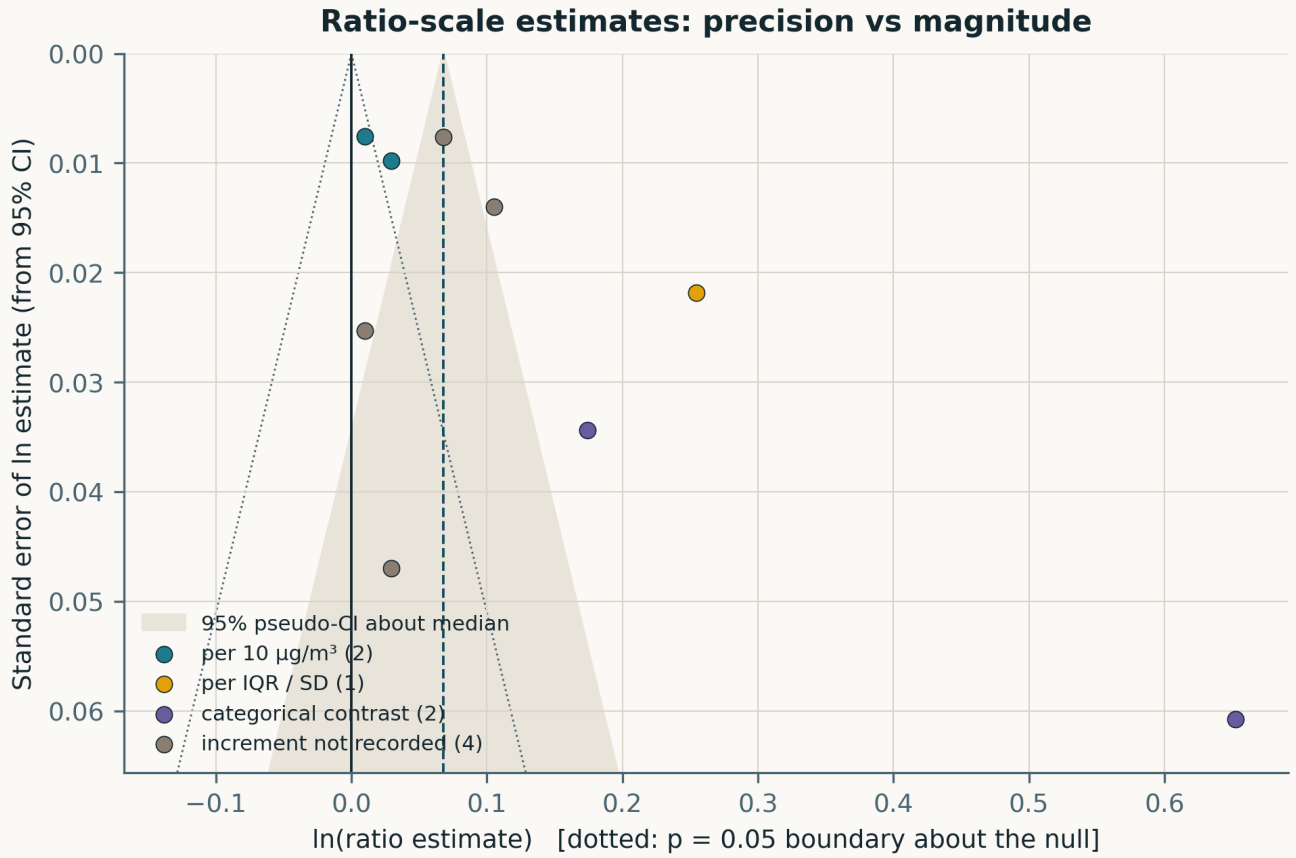


Fig. Every ratio-scale estimate (n=9) plotted as ln(estimate) against its standard error, coloured by the exposure increment the authors reported. 6 of 9 exclude the null. Increments: increment not recorded 4, per 10 $\mu g/m^3$ 2, categorical contrast 2, per IQR / SD 1. Because increments differ (per 1 unit, per 10 $\mu g/m^3$, per IQR, categorical), the spread is not a heterogeneity estimate and the median line is not a pooled effect; the plot shows whether precision and magnitude co-vary.

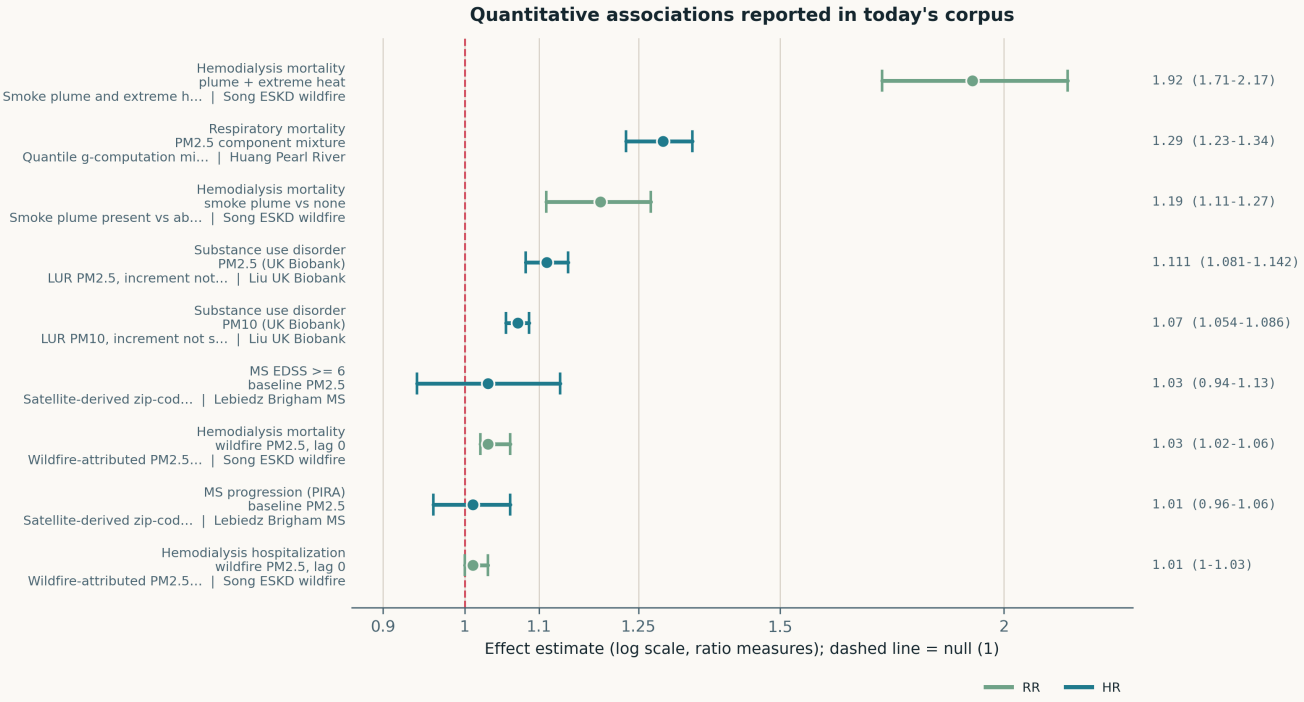


Fig. Ratio scale, 9 estimates from 4 sources: Song hemodialysis wildfire (4: two per 10 $\mu g/m^3$ wildfire $PM_{2.5}$, two categorical plume contrasts), Huang Pearl River (1, per quartile of the component mixture), Liu UK Biobank (2, increment not stated) and Lebiedz MS cohort (2, increment not stated). Four of nine estimates carry no stated increment, and the two largest are categorical contrasts; endpoints differ across sources, so no pooling and no I^2.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Neuro / mental health — 3 records

Liu Xi'an UK Biobank $PM_{2.5}$ PM_{10} PMcoarse NO_2 NOx greenness incident substance use disorder Cox 10.9 y BMC Psychiatry

Various ambient air pollutants, residential greenness, and risk of substance use disorders: a population-based cohort study.

247,138 UK Biobank participants free of substance use disorder at baseline; 4,982 incident cases over a median 10.9 years. All five pollutants were positively associated; $PM_{2.5}$ strongest (HR 1.111, 1.081-1.142), then PM_{10} (1.070, 1.054-1.086); the increment is not stated in the abstract, so the size of these HRs cannot be interpreted. Green space (HR 0.997, 0.996-0.998) was inversely associated and domestic garden attenuated the pollutant HRs in the high-exposure group. Effects were largest in older men and ever-smokers, the strata where residual confounding by deprivation and smoking intensity is worst; baseline LUR surfaces stand in for decade-long exposure. Relevance: low - an LUR cohort result without an increment adds a hazard ratio that cannot be pooled.

doi: 10.1186/s12888-026-08588-8

Lebiedz Chitnis Brigham MS Center $PM_{2.5}$ satellite zip-code PIRA EDSS 6 Cox null northeastern US 2017-2023 Brain Behav

Particulate Matter is Not Associated With Disease Worsening in Patients With Multiple Sclerosis: Retrospective Results From a Clinic-Based Cohort in the Northeastern United States.

3,040 people with MS at the Brigham MS Center, mean age 49.5, median EDSS 2.7, mean baseline $PM_{2.5}$ 5.8 $\mu g\,m^{-3}$. Adjusted HRs were null for PIRA (baseline 1.01, 0.96-1.06; time-varying 0.98, 0.93-1.04) and for EDSS >= 6 (baseline 1.03, 0.94-1.13; time-varying 1.01, 0.92-1.10). The increment behind the HRs is not stated. Exposure contrast is narrow at New England levels, zip-code assignment blurs it further, and clinic-based retrospective ascertainment of PIRA is noisy, so the null mainly bounds large effects at low concentrations. Relevance: a well-powered null at ~6 $\mu g\,m^{-3}$; resolving within-zip contrast would need denser measurement than satellite surfaces give.

doi: 10.1002/brb3.71878

Guilin critical review neurovascular glial recovery after particulate exposure reduction filtration diesel BBB microglia Neurotoxicology

Asynchronous neurovascular and glial responses after particulate air pollution exposure declines.

Critical review of experimental post-exposure studies and human exposure-reduction studies, separating cellular state, direct function and attribution. Early blood-brain barrier leakage after diesel exhaust is documented but later normalisation of the same function is not; some inflammatory markers attenuate after a switch to filtered air while other abnormalities persist for weeks or months. Human studies give both favourable and null cognitive outcomes after exposure reduction, without cell-specific recovery. The review argues for asynchronous responses rather than a common recovery sequence and sets criteria (repeated measurement of one task) for claiming recovery. Relevance: indirect - filtration-intervention trials need verified exposure drops, which indoor sensors provide.

doi: 10.1016/j.neuro.2026.103598

Respiratory & allergic — 2 records

Huang Wei Pearl River Cohort $PM_{2.5}$ components NH_4 NO_3 BC OM SO_4 Cl respiratory mortality quantile g-computation South China Environ Sci Technol Lett

Long-Term Exposure to $PM_{2.5}$ Components and Respiratory Disease Mortality in South China: The Pearl River Cohort Study

182,009 Pearl River Cohort participants; marginal structural Cox models for single components and quantile g-computation for the mixture. All six components raised respiratory mortality, single-component HRs 1.42-2.75 (increments not given in the abstract). Each quartile increase in the mixture index gave HR 1.29 (1.23-1.34), with nitrate carrying the largest weight (0.48), then organic matter and chloride. Older adults, non-smokers and non-exercisers were most susceptible (HRs 1.25-2.02). Component surfaces are model products whose errors are correlated across species, so the weight ranking partly reflects which component the model resolves best; HRs of 1.4-2.8 per component are implausibly large without the increment. Relevance: nitrate is the semi-volatile, hygroscopic fraction behind most low-cost-versus-reference disagreement in humid seasons.

doi: 10.1021/acs.estlett.6c00573

Peking University review ozone $PM_{2.5}$ heat coexposure triple exposure airway inflammation mechanisms immune-neuro-microbiota J Appl Toxicol

Mechanisms of Airway Inflammatory Diseases Under Ozone Single Exposure, Coexposure, and Triple Exposure With $PM_{2.5}$ and Heat: A Review.

Review of mechanisms linking ozone alone, ozone with $PM_{2.5}$ or heat, and all three to airway inflammation. Ozone injures epithelium oxidatively and activates NF-kB, MAPK and NLRP3; $PM_{2.5}$ is framed as a chemical amplifier giving additive or synergistic oxidative burden, whereas heat acts as a biological sensitiser through TRP-mediated neuroimmune activation and microbiota-metabolism shifts. Triple exposure is argued to overwhelm compensation through higher effective dose, lower tolerance and convergent signalling, but the authors state its health effects are largely unexplored. No quantitative synthesis and no search protocol are described. Relevance: low for sensing; co-located O_3, $PM_{2.5}$ and temperature are what multi-pollutant low-cost nodes already log.

doi: 10.1002/jat.70501

Reproductive & developmental 1 record

Shandong mouse intratracheal $PM_{2.5}$ ovary single-cell RNA-seq leukocyte infiltration granulosa cholesterol biosynthesis STAT1 atresia Toxicology

Single-cell transcriptomic profiling of $PM_{2.5}$ ovarian toxicity: pulmonary particulate instillation associates with leukocyte infiltration and impaired granulosa cholesterol biosynthesis.

Mouse model of repeated pulmonary $PM_{2.5}$ instillation with ovarian single-cell RNA-seq and in situ validation. Instillation expanded the ovarian immune compartment (B-lineage lymphocytes, neutrophils); resident macrophages took a C1qa-rich scavenger profile without NLRP3 activation. Immune-derived TGF-beta1 and TNF signalling raised JAK-STAT activity in somatic cells and suppressed the granulosa cholesterol-biosynthesis programme; trajectory modelling nominated STAT1 for follicular atresia, with p-STAT1 and TUNEL confirming in tissue. Dose, particle source and group sizes are not in the abstract, and bolus instillation is not inhalation. Relevance: low for sensing; extends the lung-to-ovary immune route behind epidemiological fecundity signals.

doi: 10.1016/j.tox.2026.154623

Mechanistic toxicology 1 record

Korea University PM_{10} brain vascular endothelial cells apoptosis DNA damage MAPK NF-kB NAC zebrafish developmental toxicity Environ Toxicol Pharmacol

Particulate Matter Induces Apoptotic Death of Brain Vascular Endothelial Cells in Association with Oxidative Damage, DNA Damage Response, and MAPK/NF-B Signaling.

PM_{10} reduced brain vascular endothelial cell viability and raised TUNEL labelling, cleaved caspase-3 and PARP, intrinsic and death-receptor apoptotic proteins, DNA-damage-response markers, MAPK/NF-kB components and lipid and protein oxidation. N-acetylcysteine pretreatment improved survival and reduced caspase-substrate cleavage, implicating redox processes. In zebrafish, PM_{10} produced concentration-dependent developmental toxicity and altered brain vascular phenotypes. The authors themselves describe the causal chain as unresolved; doses, particle provenance and whether concentrations approach realistic deposition are not given in the abstract. Relevance: low - mechanism plausibility for the cerebrovascular and neurodevelopmental associations seen in cohorts.

doi: 10.1016/j.etap.2026.105230

Exposure assessment & modelling 7 records

Anhui WRF-Chem GOCART Shao KOK dust schemes February 2026 East Asian dust storm size-resolved PM_{10} Gobi source attribution Atmosphere

Numerical Analysis of an Early-Arriving East Asian Dust Storm in 2026: Evaluation, Size-Resolved Dust Contribution, and Source Attribution

WRF-Chem case study of the 18-23 February 2026 dust storm that reached the Yangtze basin, comparing GOCART, Shao2001 and KOK emission schemes. A tuned GOCART run matched observed AOD best (R 0.74, RMSE 0.16). Size bins 5-8 supplied over 90% of dust mass in surface PM_{10}; regional emission peaked above 4.0 Tg on 20 February while area-mean loading peaked at about 90 mg/m2 a day later; the Gobi supplied about two-thirds of the loading. Validation is against AOD, not surface PM_{10} monitors, and one event cannot rank schemes. Relevance: coarse-dominated dust episodes are where $PM_{2.5}$-tuned optical sensors under-read PM_{10} most; event-specific size splits like this are what a dust correction needs.

doi: 10.3390/atmos17111107

Zhang Peking deep neural network global surface HONO reconstruction 2005-2021 megacities ozone particulate nitrate drivers Nat Commun

Long-term nitrous acid evolution modulates ozone and secondary aerosol variations in global megacities.

A deep neural network trained on worldwide field campaigns reconstructs surface HONO; Beijing fell 47.4% (2013-2021), Delhi rose 10.0% (2005-2018), and London, New York and Los Angeles declined slightly. HONO change is identified as a major driver of ozone and particulate nitrate variation, outweighing NOx and VOC abatement in Beijing and Delhi, and the framework recommends NOx control in summer and joint NOx, $PM_{2.5}$ and direct-combustion cuts in winter. HONO is sparsely measured and campaign-biased, so the reconstruction and the attribution share one learned model; no uncertainty on the nitrate share is given in the abstract. Relevance: indirect - explains nitrate-driven $PM_{2.5}$ trends that sensor networks record but cannot apportion.

doi: 10.1038/s41467-026-78349-4

NIEHS PEGS GIS-ExWAS 88 geospatial exposure indicators 11 phenotypes IPTW LASSO air pollution cardiovascular J Expo Sci Environ Epidemiol

Applying geographic information systems data linkages for an exposome-wide association study in the Personalized Environment and Genes Study

7,531 Personalized Environment and Genes Study participants' residential histories linked by open-source GIS to 88 exposure indicators; cross-sectional associations with 11 phenotypes by adjusted logistic regression, IPTW and LASSO. The GIS-ExWAS recovered associations complementary to the cohort's earlier questionnaire ExWAS, including air pollution with cardiovascular disease, plus new suggestive associations. No effect sizes are given; the contribution is the linkage pipeline. Cross-sectional design and many correlated indicators make every hit a hypothesis, and the abstract does not say which air-pollution surfaces were linked. Relevance: indirect - a standard linkage layer is where sensor-network surfaces would have to plug in.

doi: 10.1038/s41370-026-00986-0

Chi NYCU PM-bound PCDD/F PCB PCN TEQ Ulaanbaatar Beijing Taipei PMF source apportionment cancer risk dust events Environ Pollut

Spatiotemporal distribution, source apportionment and cancer risk of PM-bound PCDD/Fs, PCBs and PCNs in three East Asian cities with contrasting emissions: Ulaanbaatar, Beijing and Taipei area.

61 TSP samples in Ulaanbaatar, 24 in Beijing (22 $PM_{2.5}$) and 26 $PM_{2.5}$ in the Taipei area. $PM_{2.5}$-bound PCDD/F TEQ averaged 107 fg/m3 in Beijing against 11.5 in Taipei; Ulaanbaatar TSP reached 128. Beijing cold-season PCDD/F TEQ rose 19-fold, and during dust events Beijing TSP TEQ (76.7) exceeded Ulaanbaatar's (48.9). PMF attributed PCDD/Fs to open burning and traffic (Ulaanbaatar), waste incineration and dust (Beijing) and boilers plus incineration and traffic (Taipei). PCDD/Fs drove over 98% of carcinogenic risk, which exceeded 1e-6 in Ulaanbaatar and Beijing, but non-inhalation pathways carried 65-90% of it. Sampling periods and size cuts differ by city, limiting comparison. Relevance: indirect - mass-based sensors cannot see the toxic-equivalent fraction that drives risk here.

doi: 10.1016/j.envpol.2026.129475

Peanusaha Auvermann Texas cattle feedlot evening dust peak Hot-Dry-Windy index water balance PM_{10} surface moisture Atmos Pollut Res

A Meteorological Indicator for Particulate Matter Emissions: Adapting the Hot-Dry-Windy Index to Predict Feedlot Evening Dust Peaks

Partial abstract (Crossref). Feedlot evening dust peaks have been documented for decades but their meteorological drivers lack long-term study, and direct surface-moisture measurement is labour-intensive. The authors adapt the Hot-Dry-Windy index and couple it with precipitation in an empirical water-balance model to estimate pen-surface moisture from routine meteorology. Under the driest surface conditions the index explained 43% of the variance in evening PM_{10} (R2 0.433, r 0.658). The deposited text ends there, so the site, record length, PM_{10} instrument and performance under wetter conditions are unknown. Relevance: a cheap meteorological predictor of agricultural dust episodes, testable against low-cost PM_{10} sensors at feedlot fences.

doi: 10.1016/j.apr.2026.103257

Reatini Stafoggia Bayesian hierarchical vs random forest spatiotemporal PM_{10} $PM_{2.5}$ NO_2 prediction Italy Atmos Environ

Comparison of Bayesian hierarchical modelling and Random Forest algorithm for spatiotemporal prediction of PM_{10}, $PM_{2.5}$ and NO_2 across Italy

METADATA ONLY. Metadata only: Elsevier day-0 deposit with no abstract in Crossref at issue time; no finding is asserted. The title states a head-to-head comparison of Bayesian hierarchical modelling and random forest for national spatiotemporal prediction of PM_{10}, $PM_{2.5}$ and NO_2 across Italy; the author list (Reatini, Nobile, Stafoggia, Stortini) suggests the Italian national exposure-modelling programme. Relevance: direct if the comparison reports uncertainty calibration as well as cross-validated R2 - a Bayesian surface gives the per-cell prediction variance that guides where to add monitors, and random forest gives none without extra machinery. Recheck at the W46 abstract sweep.

doi: 10.1016/j.atmosenv.2026.122456

Saurabh Sarangi elemental characterization PM_{10} sources pristine high-altitude western Himalaya Atmos Environ

Elemental characterization of PM_{10} sources in the pristine high-altitude western Himalayan region

METADATA ONLY. Metadata only: Elsevier day-0 deposit with no abstract in Crossref at issue time; no finding is asserted. The title describes elemental characterisation and source identification of PM_{10} at a pristine high-altitude site in the western Himalaya; geography is taken from the title. Sampling method, period, analytical technique and source-apportionment approach are unknown, so whether this is a filter-based XRF or ICP campaign cannot be told. Relevance: background-site composition fixes the dust and long-range transport baseline against which Indo-Gangetic and Himalayan foothill sensor networks are interpreted. Recheck at the W46 abstract sweep.

doi: 10.1016/j.atmosenv.2026.122492

Sensing, forecasting & instrumentation — 4 records

Parthenope independent evaluation Aurora 0.4 degree air pollution foundation model CAMS persistence stations negative $PM_{2.5}$ 2024 Research Square (preprint)

Independent evaluation of Aurora air pollution forecasts beyond benchmark skill

Preprint. 96 initialisations across 2024 of the released Aurora air-pollution checkpoint, scored against CAMS, persistence and hourly background stations in three regions. Relative to persistence, Aurora cut MSE by 38.1-42.6% for $PM_{2.5}$ and 49.1-52.3% for PM_{10}, and its station MAE beat the benchmark-selected CAMS fields in all six region-pollutant pairs with bootstrap intervals excluding zero. But at +12 h, 6.27% of $PM_{2.5}$ and 7.84% of PM_{10} grid values were negative, the model gives no forecast distribution, and its response to perturbed emission inputs is untested against real emission change. Background stations only, so urban hotspots are unscored. Relevance: dense low-cost networks are the obvious test bed for exactly the near-source skill this evaluation could not reach.

doi: 10.21203/rs.3.rs-11417641/v1

Shenzhen Mini-WRAS size-resolved ultrafine particle hourly forecasting MLR RF XGBoost LSTM history window 2023-2025 Environ Pollut

Forecasting hourly size-resolved ambient ultrafine particles with size-specific historical windows.

8,982 quality-controlled hourly observations (Sep 2023-Jun 2025) from a Mini-WRAS at an urban Shenzhen site; 270 model-window-target configurations (MLR, RF, XGBoost, LSTM, BiLSTM; 1-24 h histories) scored on the same 533 test hours. Best NRMSE was 60.2% at 10 nm and 22.1% at 14 nm against 13.4-14.8% at 19-100 nm; a 2 h history was best for 10-19 nm and 4 h for 27-100 nm, while the summed UFP target reached 13.7% at 12 h. Several model classes tied, so architecture mattered less than window. One site, one-step-ahead, and no meteorological or traffic covariates are described. Relevance: aggregate number forecasts hide the poorly predictable nucleation mode, which is also the size range low-cost counters miss.

doi: 10.1016/j.envpol.2026.129537

Tohidian Elia denoising neural processes input uncertainty $PM_{2.5}$ spatial interpolation air quality sensors 2026 arXiv (preprint)

DnNP: Denoising Input Uncertainty in Neural Processes

Preprint. Denoising Neural Processes extend the neural-process meta-learner to treat inputs as noisy, trained by variational inference, with a proof that the Bayes-optimal predictor under input noise lies in the model class and that fixed-variance plug-in models carry an irreducible KL gap. On $PM_{2.5}$ spatial interpolation from real-world air-quality sensors the method cut RMSE by up to 30% and NLL by up to 73% against standard neural processes. The abstract names neither the network, its sensor type nor the baseline beyond vanilla NPs, so 'real-world sensors' cannot be placed against kriging or calibrated-network practice. Relevance: modelling noisy and mis-located sensor inputs explicitly is exactly the low-cost network problem; worth reading if the PM data are PurpleAir-class.

doi: 10.48550/arXiv.2611.14424

Punjab ARIMA LSTM NARX forecasting Deep Blue AOD550 Thar Thal Karakum Kutch Kyzylkum Arabian deserts 2000-2030 Environ Monit Assess

Forecasting of aerosol optical depth over arid and semi-arid regions using time series and machine learning models.

294 monthly Deep Blue AOD550 values (Feb 2000-Jul 2024) for the Thar, Thal, Karakum, Kutch, Kyzylkum and Arabian deserts; ARIMA, LSTM and NARX evaluated on a 44-month test set. NARX was best (test RMSE 0.0180-0.0275) over LSTM (0.0215-0.0350, R2 0.53-0.70) and ARIMA (0.0250-0.0400, R2 0.40-0.60), and was used to project AOD to July 2030. Monthly means of a column optical quantity carry no surface PM information, and a six-year recursive forecast from 250 training points has no credible uncertainty statement. Relevance: marginal - desert dust is the regime where AOD-to-$PM_{2.5}$ conversion and optical low-cost sensors both fail, but this paper does not touch the surface.

doi: 10.1007/s10661-026-16140-0

Burden, policy & mitigation 5 records

CUHK CESM XGBoost 0.1 degree $PM_{2.5}$ mortality GEMM SSP1 SSP2 SSP3 SSP5 2010 2050 emissions climate land use GeoHealth

High-Resolution $PM_{2.5}$ Mortality Projections With Machine Learning Reveal Global Inequities Under Future Climate and Socioeconomic Pathways

CESM factorial runs separating emissions, climate and land-use change, bias-corrected and refined to 0.1 degree with XGBoost, fed to GEMM. Present-day burden about 6.13 million deaths in 2010 (UI 5.77-6.48); by 2050 SSP1 removes about 1.1 million deaths and SSP3 adds about 0.5 million. Emissions dominate, climate change offsets part of the emission benefit, and the burden concentrates in people over 60 and in LMICs. The uncertainty interval covers the concentration-response function only; the ML downscaler is trained against present-day observations and is applied to 2050 states it has never seen. Relevance: the downscaler's skill rests on ground data that are thinnest in the LMICs that carry the burden.

doi: 10.1029/2026gh001829

Shandong GBD 2021 $PM_{2.5}$ attributable ASMR greenspace access Gini 204 countries 2000-2020 EAPC BMC Public Health

Inequalities in greenspace access and their associations with all-cause and cause-specific mortality attributable to $PM_{2.5}$: a global analysis from 2000 to 2020.

204 countries, 2000-2020. Global $PM_{2.5}$-attributable all-cause ASMR fell from 63.5 to 55.4 per 100,000 while greenspace access fell from 37.0% to 28.6%; greenspace inequality in low-SDI countries rose (Gini 0.20 to 0.24). Higher greenspace access correlated with lower $PM_{2.5}$-attributable mortality for IHD, stroke, COPD and neonatal disorders. Both sides are modelled country aggregates and the attributable burden is itself a function of GBD exposure and baseline rates, so the correlation is ecological and confounded by development. Mortality attributable to $PM_{2.5}$ can fall while concentrations rise if baseline rates fall, so the trend is not an exposure trend. Relevance: low - burden bookkeeping with no measurement content.

doi: 10.1186/s12889-026-28654-w

Luebeck Schleswig-Holstein emergency department climate air quality infectious surveillance DLNM register protocol 2014-2024 JMIR Res Protoc

Climate, Air Quality, and Emergency Department Use in Northern Germany: Protocol for a Multicenter Register Study.

Protocol for a four-site retrospective study of emergency department presentations in the Luebeck area, January 2014-December 2024, linking ICD-coded visits to German Weather Service meteorology, state air-quality monitoring and Robert Koch Institute infectious surveillance. Distributed lag nonlinear models within GAMs, stratified by age, sex and socioeconomic status; ethics granted July 2025 and data transfer under way. No results. The analysis plan centres on temperature; the abstract names no particle metric, and a maritime region with few monitors will give little PM contrast. Relevance: low - noted as an upcoming low-concentration ED series.

doi: 10.2196/92900

Liu Guanzhong Basin agricultural reactive nitrogen ammonia emissions air quality impact Atmos Pollut Res

Impacts of agricultural reactive nitrogen emissions on the air quality in the Guanzhong Basin, China

METADATA ONLY. Metadata only: Elsevier day-0 deposit with no abstract in Crossref at issue time; no finding is asserted. The title addresses the impact of agricultural reactive nitrogen emissions on air quality in the Guanzhong Basin (Xi'an region); whether $PM_{2.5}$ is the endpoint is not stated in the metadata, and it is carried because agricultural ammonia is the lever on secondary ammonium nitrate in this basin. Relevance: ties to today's nitrate theme (Pearl River components, HONO reconstruction) if the abstract confirms a $PM_{2.5}$ outcome; recheck at the W46 abstract sweep. Model or observation basis is unknown.

doi: 10.1016/j.apr.2026.103267

Li Cao plant communities relative reduction efficiency atmospheric microplastics $PM_{2.5}$ urban green space Atmos Pollut Res

Assessing the relative reduction efficiency of different plant communities to atmospheric microplastics and $PM_{2.5}$

METADATA ONLY. Metadata only: Elsevier day-0 deposit with no abstract in Crossref at issue time; no finding is asserted. The title compares how different plant communities reduce atmospheric microplastics and $PM_{2.5}$, presumably by paired in- and out-of-canopy measurement, but neither design, site, season nor instrument is available, and microplastic counting methods are not standardised. Relevance: vegetation-barrier claims rest on short paired campaigns where sensor inter-unit bias is the same size as the effect; worth checking how the reduction was measured, and over how many paired days, once the abstract appears.

doi: 10.1016/j.apr.2026.103268

Other clinical endpoints 3 records

Song Fresenius hemodialysis ESKD wildfire smoke plume $PM_{2.5}$ extreme heat case-crossover mortality West US 2010-2018 Environ Epidemiol

Exposure to wildfire smoke and risk of mortality and hospitalization among hemodialysis patients in the Western United States: Effect modification by extreme heat events.

55,223 adult hemodialysis patients at Fresenius clinics in California, Oregon and Washington, warm seasons 2010-2018; conditional quasi-Poisson. A smoke plume overhead raised same-day mortality (RR 1.19, 1.11-1.27); each 10 $\mu g\,m^{-3}$ of wildfire $PM_{2.5}$ raised mortality 3% (RR 1.03, 1.02-1.06) and hospitalization 1% (1.01, 1.00-1.03). Plume plus extreme heat gave RR 1.92 (1.71-2.17) against 1.02 (0.95-1.10) for plume alone, so the plume signal is mostly heat-coincident. Exposure is assigned to the clinic, not the residence, and wildfire $PM_{2.5}$ is a modelled split of total $PM_{2.5}$; the interaction contrast is large enough that residual confounding by heat timing remains the main threat. Relevance: a dialysis population attends clinics thrice weekly, which makes clinic-sited low-cost monitors a cheap, exact exposure point.

doi: 10.1097/ee9.0000000000000541

Yan Li Sun Yat-sen national cohort $PM_{2.5}$ components OM BC SO4 NO_3 NH4 arthritis quantile g-computation middle-aged older China GeoHealth

Component-Specific $PM_{2.5}$ Exposure and Arthritis Risk in China: Evidence of Differential Susceptibility in a National Cohort

14,935 middle-aged and older adults in a nationally representative Chinese cohort; mixed-effects logistic regression and quantile g-computation. Both short- and long-term $PM_{2.5}$ mass and organic matter, black carbon, sulfate and ammonium were reported to raise arthritis risk, with men and overweight or obese participants most vulnerable. The abstract gives no effect size, CI or increment, and does not say how arthritis was ascertained, so neither magnitude nor outcome specificity can be judged from the record. Component surfaces are correlated model products. Relevance: low until numbers appear; another component study where nitrate does not lead, unlike the Pearl River respiratory result today.

doi: 10.1029/2025gh001702

Guangzhou Tianjin Guizhou urolithiasis stone composition NO_2 PM_{10} inverse U-shaped case-crossover 128000 stones BMC Public Health

Ambient nitrogen dioxide and composition-specific risk of urolithiasis: a multicenter time-stratified case-crossover study.

25,374 analysed urinary stones in Guangzhou with validation in Tianjin (99,552) and Guizhou (3,021); conditional logistic regression and GAMs over lags 0-7 and 6- to 12-month averages. The robust signal is NO_2, strongest at lags 0 and 2-4 for carbonate apatite and calcium oxalate. PM_{10} showed mainly inverse or U-shaped associations with overall and composition-specific risk, and a PM_{10}-carbonate apatite pattern was confirmed in Guizhou; no PM effect sizes are given in the abstract. Stones are dated at analysis, not symptom onset, so short lags are misaligned with formation. Relevance: low - a PM null-to-inverse result in a gas-driven outcome, carried for completeness.

doi: 10.1186/s12889-026-28561-0

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Guilin critical review neurovascular glial recovery after particulate exposure reduction filtration diesel BBB microglia	Asynchronous neurovascular and glial responses after particulate air pollution exposure declines.	*Neurotoxicology*	Critical review	Particulate air pollution after exposure decline (filtration, emission control); diesel exhaust models	Global / multi-region
Lebiedz Chitnis Brigham MS Center $PM_{2.5}$ satellite zip-code PIRA EDSS 6 Cox null northeastern US 2017-2023	Particulate Matter is Not Associated With Disease Worsening in Patients With Multiple Sclerosis: Retrospective Results From a Clinic-Based Cohort in the Northeastern United States.	*Brain Behav*	Retrospective cohort	Monthly zip-code $PM_{2.5}$ from an open satellite-derived dataset, baseline and time-varying annual means	USA
Liu Xi'an UK Biobank $PM_{2.5}$ PM_{10} PM_{coarse} NO2 NOx greenness incident substance use disorder Cox 10.9 y	Various ambient air pollutants, residential greenness, and risk of substance use disorders: a population-based cohort study.	*BMC Psychiatry*	Prospective cohort	Residential $PM_{2.5}$, $PM_{2.5-10}$, PM_{10}, NO2, NOx from land-use regression; 300 m greenness	UK
Reproductive & developmental					
Shandong mouse intratracheal $PM_{2.5}$ ovary single-cell RNA-seq leukocyte infiltration granulosa cholesterol biosynthesis STAT1 atresia	Single-cell transcriptomic profiling of $PM_{2.5}$ ovarian toxicity: pulmonary particulate instillation associates with leukocyte infiltration and impaired granulosa cholesterol biosynthesis.	*Toxicology*	Animal experiment	Repeated pulmonary $PM_{2.5}$ instillation in mice (dose not stated in abstract)	China
Respiratory & allergic					
Huang Wei Pearl River Cohort $PM_{2.5}$ components NH4 NO3 BC OM SO4 Cl respiratory mortality quantile g-computation South China	Long-Term Exposure to $PM_{2.5}$ Components and Respiratory Disease Mortality in South China: The Pearl River Cohort Study	*Environ Sci Technol Lett*	Prospective cohort	Long-term $PM_{2.5}$ components (NH4+, NO3-, BC, OM, SO4 2-, Cl-), modelled surfaces	China
Peking University review ozone $PM_{2.5}$ heat coexposure triple exposure airway inflammation mechanisms immune-neuro-microbiota	Mechanisms of Airway Inflammatory Diseases Under Ozone Single Exposure, Coexposure, and Triple Exposure With $PM_{2.5}$ and Heat: A Review.	*J Appl Toxicol*	Narrative review	Ozone with $PM_{2.5}$ and heat coexposure (mechanistic literature)	Global / multi-region
Mechanistic toxicology					
Korea University PM_{10} brain vascular endothelial cells apoptosis DNA damage MAPK NF-kB NAC zebrafish developmental toxicity	Particulate Matter Induces Apoptotic Death of Brain Vascular Endothelial Cells in Association with Oxidative Damage, DNA Damage Response, and MAPK/NF-B Signaling.	*Environ Toxicol Pharmacol*	Animal + in vitro	PM_{10} (source and dose not stated in abstract)	South Korea
Sensing, forecasting & instrumentation					
Parthenope independent evaluation Aurora 0.4 degree air pollution foundation model CAMS persistence stations negative $PM_{2.5}$	Independent evaluation of Aurora air pollution forecasts beyond benchmark skill	*Research Square (preprint)*	Benchmark / forecasting model comparison	Forecast $PM_{2.5}$ and PM_{10} (Aurora foundation model, 0.4 degree) against CAMS and hourly background stations	Global / multi-region
Punjab ARIMA LSTM NARX forecasting Deep Blue AOD550 Thar Thal Karakum Kutch Kyzylkum Arabian deserts 2000-2030	Forecasting of aerosol optical depth over arid and semi-arid regions using time series and machine learning models.	*Environ Monit Assess*	Benchmark / forecasting model comparison	Monthly satellite AOD550 (MODIS Deep Blue), six Asian deserts	Asia
Shenzhen Mini-WRAS size-resolved ultrafine particle hourly forecasting MLR RF XGBoost LSTM history window 2023-2025	Forecasting hourly size-resolved ambient ultrafine particles with size-specific historical windows.	*Environ Pollut*	Benchmark / forecasting model comparison	Ultrafine particle number in eight bins 10-100 nm, Mini-WRAS, one urban site	China

First author	Focus	Journal	Design	Metric	Region
Tohidian Elia denoising neural processes input uncertainty $PM_{2.5}$ spatial interpolation air quality sensors	DnNP: Denoising Input Uncertainty in Neural Processes	*arXiv (preprint)*	Method / algorithm	$PM_{2.5}$ from real-world air-quality sensor networks used as a spatial-interpolation benchmark	Not stated
Exposure assessment & modelling					
Anhui WRF-Chem GOCART Shao KOK dust schemes February 2026 East Asian dust storm size-resolved PM_{10} Gobi source attribution	Numerical Analysis of an Early-Arriving East Asian Dust Storm in 2026: Evaluation, Size-Resolved Dust Contribution, and Source Attribution	*Atmosphere*	Chemical transport model	Size-resolved mineral dust contribution to surface PM_{10} and AOD, WRF-Chem with three emission schemes	China
Chi NYCU PM-bound PCDD/F PCB PCN TEQ Ulaanbaatar Beijing Taipei PMF source apportionment cancer risk dust events	Spatiotemporal distribution, source apportionment and cancer risk of PM-bound PCDD/Fs, PCBs and PCNs in three East Asian cities with contrasting emissions: Ulaanbaatar, Beijing and Taipei area.	*Environ Pollut*	Intensive field campaign, PMF source apportionment	$PM_{2.5}$-bound (Beijing, Taipei) and TSP-bound (Ulaanbaatar) PCDD/Fs, PCBs, PCNs; filter sampling	Asia
NIEHS PEGS GIS-ExWAS 88 geospatial exposure indicators 11 phenotypes IPTW LASSO air pollution cardiovascular	Applying geographic information systems data linkages for an exposome-wide association study in the Personalized Environment and Genes Study	*J Expo Sci Environ Epidemiol*	Exposome-wide association	88 GIS-linked residential exposure indicators including air pollution	USA
Peanusaha Auvermann Texas cattle feedlot evening dust peak Hot-Dry-Windy index water balance PM_{10} surface moisture	A Meteorological Indicator for Particulate Matter Emissions: Adapting the Hot-Dry-Windy Index to Predict Feedlot Evening Dust Peaks	*Atmos Pollut Res*	Statistical analysis of monitoring data	Evening PM_{10} concentrations downwind of a cattle feedlot	USA
Reatini Stafoggia Bayesian hierarchical vs random forest spatiotemporal PM_{10} $PM_{2.5}$ NO2 prediction Italy	Comparison of Bayesian hierarchical modelling and Random Forest algorithm for spatiotemporal prediction of PM_{10}, $PM_{2.5}$ and NO2 across Italy	*Atmos Environ*	Metadata only	Spatiotemporal PM_{10}, $PM_{2.5}$ and NO2 prediction surfaces	Italy
Saurabh Sarangi elemental characterization PM_{10} sources pristine high-altitude western Himalaya	Elemental characterization of PM_{10} sources in the pristine high-altitude western Himalayan region	*Atmos Environ*	Metadata only	PM_{10} elemental composition and sources	India
Zhang Peking deep neural network global surface HONO reconstruction 2005-2021 megacities ozone particulate nitrate drivers	Long-term nitrous acid evolution modulates ozone and secondary aerosol variations in global megacities.	*Nat Commun*	Data-fusion / ML surrogate model	Secondary $PM_{2.5}$ (particulate nitrate) and ozone driven by reconstructed surface HONO	Global / multi-region
Burden, policy & mitigation					
CUHK CESM XGBoost 0.1 degree $PM_{2.5}$ mortality GEMM SSP1 SSP2 SSP3 SSP5 2010 2050 emissions climate land use	High-Resolution $PM_{2.5}$ Mortality Projections With Machine Learning Reveal Global Inequities Under Future Climate and Socioeconomic Pathways	*GeoHealth*	Integrated emission-air quality-health modelling	Annual $PM_{2.5}$ from CESM bias-corrected and downscaled to 0.1 degree by XGBoost	Global / multi-region
Li Cao plant communities relative reduction efficiency atmospheric microplastics $PM_{2.5}$ urban green space	Assessing the relative reduction efficiency of different plant communities to atmospheric microplastics and $PM_{2.5}$	*Atmos Pollut Res*	Metadata only	Atmospheric $PM_{2.5}$ and airborne microplastics under different plant communities	Not stated
Liu Guanzhong Basin agricultural reactive nitrogen ammonia emissions air quality impact	Impacts of agricultural reactive nitrogen emissions on the air quality in the Guanzhong Basin, China	*Atmos Pollut Res*	Metadata only	Air-quality response to agricultural reactive nitrogen emissions ($PM_{2.5}$ likely via ammonium nitrate; not confirmed)	China
Luebeck Schleswig-Holstein emergency department climate air quality infectious surveillance DLNM register protocol 2014-2024	Climate, Air Quality, and Emergency Department Use in Northern Germany: Protocol for a Multicenter Register Study.	*JMIR Res Protoc*	Study protocol	State environmental monitoring air-quality data linked with German Weather Service meteorology	Germany

First author	Focus	Journal	Design	Metric	Region
Shandong GBD 2021 $PM_{2.5}$ attributable ASMR greenspace access Gini 204 countries 2000-2020 EAPC	Inequalities in greenspace access and their associations with all-cause and cause-specific mortality attributable to $PM_{2.5}$: a global analysis from 2000 to 2020.	*BMC Public Health*	Descriptive burden analysis, GBD 2021	GBD 2021 ambient $PM_{2.5}$-attributable age-standardised mortality; greenspace access from GLAD	Global / multi-region
Other clinical endpoints					
Guangzhou Tianjin Guizhou urolithiasis stone composition NO2 PM_{10} inverse U-shaped case-crossover 128000 stones	Ambient nitrogen dioxide and composition-specific risk of urolithiasis: a multicenter time-stratified case-crossover study.	*BMC Public Health*	Time-stratified case-crossover	Six criteria pollutants incl. $PM_{2.5}$ and PM_{10} (fixed-site monitors) with meteorology	China
Song Fresenius hemodialysis ESKD wildfire smoke plume $PM_{2.5}$ extreme heat case-crossover mortality West US 2010-2018	Exposure to wildfire smoke and risk of mortality and hospitalization among hemodialysis patients in the Western United States: Effect modification by extreme heat events.	*Environ Epidemiol*	Time-stratified case-crossover	Smoke-plume presence (satellite plume polygons) and wildfire-attributed $PM_{2.5}$ at each dialysis facility; extreme heat events	USA
Yan Li Sun Yat-sen national cohort $PM_{2.5}$ components OM BC SO4 NO3 NH4 arthritis quantile g-computation middle-aged older China	Component-Specific $PM_{2.5}$ Exposure and Arthritis Risk in China: Evidence of Differential Susceptibility in a National Cohort	*GeoHealth*	Prospective cohort	Short- and long-term $PM_{2.5}$ mass and components (OM, BC, SO4 2-, NO3-, NH4+), modelled surfaces	China

Provenance and method

Entry window **10 November 2026**, single day (Tuesday). Harvester legs: **PubMed** health **25**, sensing **3**; **Europe PMC 26**; **Crossref by ISSN 64**; **arXiv** 1 in-window record (carried); **OpenAlex** HTTP 429. 119 raw, 111 unique, 17 already seen → 94 fresh. PubMed connector (EDAT 10–11 November, three axes) 60 PMIDs, 27 bucketed to 10 November: 6 seen, 1 PMID merged onto a harvested record, 10 appended → 104; the cross-day pass removed 12 previously seen or rejected → **92 screened**. Abstracts: efetch filled 15 of 16 PMID records; Crossref recovered a partial abstract for 1 of 5 Elsevier records tried. OpenAlex, alphaXiv, bioRxiv/medRxiv and Wiley Scholar Gateway connectors were not available on this session; **Scite** retraction check not run. No Amass or Consensus backfill this run. **26 admitted**, **66 rejected** with logged reasons. **4 carried on metadata alone**. Affiliations from PubMed XML and Crossref (OpenAlex unavailable). Trial watch: see W46 weekly. Abstracts remain the copyright of their publishers.